

As technology reshapes the world around us, robotics is opening exciting career pathways for young minds ready to make an impact. From smart machines to automated systems, robotics is driving innovation across industries—from healthcare and mining to space and manufacturing.

Now is the perfect time to gain practical, future-focused skills in a field that's growing fast and full of opportunity. Whether you love solving problems with your hands or designing high-tech systems, robotics offers rewarding careers that combine creativity, challenge, and purpose.

ROBOTICS & AUTOMATION

PATHWAYS LEGEND

- Primary Pathway
- Optional Pathways
- ▶ Entry Points No Pre-Requisite
- ▶ Entry with Recommendations See Website
- * Free or Low Fee

INTRO

- * ○ Certificate II in Integrated Technologies (Robotics Control Stream) 22586VIC +VETDSS*
- RII21222 Certificate II in Autonomous Workplace Operations VETDSS*

NEXT STEP - PROGRESS to one of these CERTIFICATE III COURSES

HIGHER SKILLS

- * ○ Programmable Logic Controller (PLC) Based Industrial Robotics Skill Set
- * ○ Industrial Robotics and IoT Coding Skill Set
- * ○ ICT30120 Certificate III in Information Technology [BFF7]
- * ○ MEM30219 [AD57] or MEM31922 [AE16] Certificate III in Engineering Apprenticeship

NEXT STEP - PROGRESS to one of these CERTIFICATE IV COURSES

PROGRESS

RELATED INDUSTRIES

Robotics and automation technologies are poised for use, in a wide range of industries in the near future.

An example is the Australian space industry which is a multidisciplinary domain that needs a broad range of people with diverse skill sets to help it grow and transform.

Other industries which are important to the future of West Australia and are set to be supported by emerging technologies in robotics are:

- Construction
- Mining
- Health
- Defence
- Education and Training
- Emergency Services
- Logistics and Transport
- Agriculture and Agribusiness
- Energy

ADVANCED SKILLS

GENERAL ROBOTICS, MECHATRONICS & AUTOMATION

- * ▶ 22697VIC CIV Integrated Technologies (Robotics)

PROGRESS

INSTRUMENTATION CONTROL & AUTOMATION

- * ○ UEE40420 Certificate IV in Electrical - Instrumentation Pre Req UEE30820
- OR
- * ▶ UEE40720 Certificate IV in Electronics and Communications (18 months)

PROGRESS

ROBOTICS SOFTWARE ENGINEERING

- * ▶ ICT40120 Certificate IV in Information Technology (AE09 Games and Intelligent Systems)

PROGRESS

ADVANCED INDUSTRIAL MANUFACTURING

- * ○ MEM40119 Certificate IV in Engineering, Apprenticeship
- * ○ CNC Machining [AD26] Boilermaking [BEJ8]
- Industrial Robotics Skill Set

PROGRESS

JOB OPTIONS

Service Technician
Installation Technician
Robotics Technician

- * ○ UEE50520 Diploma of Electronics and Communications Engineering [BFP5]

PROGRESS

JOB OPTIONS

Mechatronics Technician
Industrial Designer
Engineering Innovation Manager
Control System Automation Technician

- * ○ ICT50220 Diploma of Information Technology (AD90 Games and Intelligent Systems)

PROGRESS

JOB OPTIONS

IoT/robotics specialist
Games and VR developer
Digital Programmer
Software Developer

JOB OPTIONS

Robotic Welding
Additive Manufacturing (3D Printing)
Subtractive Manufacturing (CnC)

MICRO LEARNING

Bite size learning opportunities to support your career

- Intro to Cyber Security Skill Set
- CAD and 3D Printing Skill Set
- Artificial Intelligence Skill Set
- Learn to Fly Drones Short Course

FURTHER STUDIES AT TAFE

- ICT50220 Diploma of Information Technology (Back End OR Front End Development)
- ICT40120 Certificate IV in Information Technology (Programming)
- AVI30419 Certificate III in Aviation (Remote Pilot)

GO TO UNIVERSITY

A North Metropolitan TAFE Diploma qualifies you for entry into many undergraduate degrees at all WA universities. Some roles like automation technician or integration engineer may not require a full engineering degree but benefit from certifications and programming skills.